

Arya Hosseini

Montreal, QC, Canada

[Portfolio](#) , [GitHub](#)

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SUMMARY

Game Programmer experienced in building **gameplay systems, tools, and engine features** in **Unity** and **Unreal Engine** using **C#** and **C++**. Strong foundation in **game architecture, systems design, and optimization**. Looking for opportunities to craft immersive, high performance gameplay experiences.

SKILLS

Engines & Tools: Unity, Unreal Engine, SFML, ImGui, Git, Shader Graph, Maya

Programming Languages: C#, C++ , Lua, SQL, JavaScript, HTML/CSS

Languages: English, French

PROFESSIONAL EXPERIENCE

Unity Gameplay Programmer Intern - Baobab Games

May 2024 - Aug 2024

DION – Multiplayer Survival

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- Expanded the **construction system**, integrating **18 modular parts** with unique constraints to enable rapid content iteration without engineering dependency.
- Built a **C# interface based interaction framework**, allowing scalable, reusable gameplay interactions across multiple systems.
- Implemented a **network synced storage system** using **Mirror Networking**, integrated with UI and inventory systems for smoother multiplayer synchronization.
- Created **custom Unity Editor tools**, cutting level setup time and enhancing designer workflow efficiency.
- Collaborated with **artists and QA** in an **Agile** environment, solving gameplay issues and ensuring feature stability.

PERSONAL PROJECT

Action Roguelike (Unity)

[View Project](#)

- Designed a **modular player architecture** using the **State Pattern**, enabling scalable gameplay and clean system separation.
- Developed a **procedural world generation** system with **zone based architecture** and **binary space partitioning**, improving replayability and runtime performance.
- Built a **data driven ability system** with **ScriptableObjects**, supporting composable abilities, randomized upgrades, and seamless UI integration.
- Created a **custom Flow Field Pathfinding system**, replacing Unity's NavMesh for efficient multi agent navigation in procedural environments.
- Implemented a **FSM based Enemy AI framework** using ScriptableObjects for reusable and scalable behaviors.
- Profiled and optimized** CPU/GPU performance using Unity Profiler, reducing overhead through refactoring, memory management, and batching.

Mini ECS Game Engine (C++)

[View Project](#)

- Built a **small 2D game engine** from scratch with **C++**, **SFML**, and **ImGui**, focusing on modular engine architecture and real time update loops.
- Implemented a **custom Entity Component System (ECS)** for efficient and data driven object management.
- Developed rendering, input, collision, and debugging systems while applying **software architecture** and **memory management** principles.

EDUCATION

DEC – Computer Science, Video Game Programming

2022 - 2024

LaSalle College

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